INVENTORY MANAGEMENT SYSTEM

User Manual

Complete guide to managing products, sales, stock, and reports

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Introduction

The Inventory Management System is a desktop application built to help businesses efficiently track their products, manage stock levels, record sales, and analyze performance data. Version 3.0 adds Cycle Count, Audit Trail, and Auto Backup to give you complete operational control. This manual walks you through each module so you can get up and running quickly.

What This System Can Do

- Add, edit, and delete products with pricing and stock information
- Record sales and automatically deduct stock
- Restock products and track all stock movements
- Handle customer returns (Sales Return) and supplier returns (Purchase Return)
- Record product losses due to damage, expiry, or spoilage
- Generate detailed profit reports with charts
- Export reports to professional PDF files
- Perform Cycle Counts — compare physical stock against system records to identify variances
- View a full Audit Trail — every action ever performed in the system is logged
- Auto Backup — data is automatically backed up every 7 days with manual trigger option

How to Navigate

The application is divided into sections accessible from the left-side navigation menu. The menu is organized into four groups: INVENTORY, OPERATIONS, ANALYTICS, and SYSTEM. Click any menu item to switch to that section. The current section loads in the main content area on the right.

> **Tip:** The left navigation menu is always visible. You can switch between sections at any time without losing data that has already been saved.

Currency & Format

All prices are displayed in Philippine Peso (₱). Dates follow MM/DD format in displays and YYYY-MM-DD internally.

Module 1

Products

View and search your entire product catalog

What It Does

The Products module displays your complete product list in a table. From here you can search, filter by category, and quickly see the stock status of every product at a glance.

Reading the Product Table

Column	Description
ID	Unique number automatically assigned to each product.
Name	The product name.
Category	Product group (e.g., Snacks, Beverages, Garments).
Unit Cost	How much you paid per unit — used for profit calculation.
Price	The selling price shown to customers.
Stock	Current quantity available in inventory.
Min Stock	The minimum level before a low-stock alert is triggered.
Status	Green 'In Stock' if stock is above minimum. Red 'Low Stock' if at or below minimum.

Searching and Filtering

Use the Search box at the top to find products by name, category, or ID. Start typing and results filter in real time.

Use the Category dropdown to show only products from a specific category. Both filters work together — you can search within a filtered category.

> **Tip:** The counter below the search bar shows 'Showing X of Y products' so you always know how many results match your filters.

Click Clear Filters to reset both the search text and category filter at once.

Low Stock Alerts

When the application starts, it automatically checks for any products where the current stock is at or below the minimum stock level. If any are found, a warning dialog appears listing all affected products.

> **Note:** Low-stock products are highlighted in red in the Status column so they remain visible even after dismissing the startup alert.

Module 2

➕Add / Edit Product

Create new products or update existing ones

Adding a New Product

To create a new product, make sure no product is selected in the list on the right, fill in the form on the left, and click Save Product.

1	Click 'Add/Edit Product' in the left navigation menu.
2	Leave the product list on the right with nothing selected (or click Clear to deselect).
3	Enter the Product Name — this field is required.
4	Select or type a Category. You can choose from the predefined list or type a custom one.
5	Enter the Unit Cost — what you paid per unit. This is used to calculate profit in reports.
6	Enter the Price — the selling price that will be charged to customers.
7	Enter the Stock Quantity — how many units are currently available.
8	Set the Minimum Stock Level — the system will alert you when stock falls to this number.
9	Click 'Save Product'. A success message confirms the product has been saved.

Editing an Existing Product

Select any product from the list on the right side. Its details will automatically populate the form on the left for you to modify.

1	Click 'Add/Edit Product' from the navigation menu.
2	Click on the product you want to edit in the right-side list.
3	The form on the left fills in with the product's current details.
4	Make your changes to any fields you need to update.
5	Click 'Save Product' to apply the changes.

> **Tip:** You can update the Unit Cost at any time. Note that reports use the current Unit Cost from the product record when calculating profit — historical sales are not retroactively recalculated.

Deleting a Product

Select a product from the list, then click the Delete button. You will be asked to confirm before the product is permanently removed.

> **Warning:** Deleting a product removes it from the product list permanently. However, existing sales and stock log records that reference the product are preserved in the data files.

Product Fields Reference

- Product Name — Required. The name displayed throughout the system.
- Category — Helps organize and filter products. Choose from the list or type a custom one.
- Unit Cost — Your cost per unit (purchase price). Must be set for profit reports to be accurate.
- Price — The retail selling price.
- Stock Quantity — Initial or current stock level.
- Minimum Stock Level — Threshold for low-stock alerts. Default is 10.

Module 3

Sell Product

Record sales and automatically update inventory

How It Works

When a sale is completed, the system records the transaction in the sales log and automatically deducts the sold quantity from the product's stock. You cannot sell more than the available stock.

Recording a Sale

1	Click 'Sell Product' in the navigation menu.
2	Select the product being sold from the dropdown list.
3	Review the product information panel that appears — it shows the current Price and Available Stock.
4	Enter the Quantity to sell.
5	Check the Transaction Summary at the bottom — it shows the calculated total automatically.
6	Click 'Complete Sale' to finalize the transaction.
7	A confirmation message appears. The product's stock is immediately updated.

Important: If the entered quantity exceeds available stock, the system will show an error and block the sale. Adjust the quantity to match what is available.

What Gets Recorded

- A sale record is saved with the product name, quantity, price, total, and timestamp.
- A stock log entry is created with action type 'Sale', showing stock before and after.
- The product's stock count is permanently reduced.

Tip: The Transaction Summary updates automatically as you change the quantity — no need to calculate the total yourself.

Module 4

Restock

Add inventory when new stock arrives

How It Works

Use the Restock module when new merchandise arrives. It increases the product's stock and records a Restock entry in the stock log for full traceability.

Restocking a Product

1	Click 'Restock' in the navigation menu.
2	Select the product you are restocking from the dropdown.
3	Review the Current Stock Information panel showing the existing stock level.
4	If the product is flagged as Low Stock, a red warning banner will appear at the top.
5	Enter the Quantity to Add.
6	The 'After Restock' preview shows what the new stock level will be.
7	Click 'Add Stock' to confirm. The stock is updated immediately.

Tip: The 'After Restock' preview helps you avoid over-ordering by showing exactly what the new level will be before you commit.

Module 5

Sales Return

Process items returned by customers

What Is a Sales Return?

A Sales Return happens when a customer brings back an item they purchased. The product is returned to your inventory and the stock count increases. This is different from a Purchase Return, which is about returning items to your supplier.

Processing a Sales Return

1	Click 'Sales Return' in the navigation menu.
2	Select the product being returned from the dropdown.
3	Verify the product details in the information panel (current price and stock).
4	Enter the Return Quantity — how many units the customer is returning.
5	Enter the Return Reason (e.g., 'Defective item', 'Wrong size', 'Customer changed mind').
6	Click 'Process Return'. The stock is increased by the returned quantity.

Note: A reason is required to process the return. This ensures your stock log has a clear explanation for every stock change.

What Gets Recorded

- A stock log entry is created with action type 'Sales Return'.
- The reason you entered is saved alongside the log entry.
- Stock before and after the return are both recorded for audit purposes.

Module 6

Purchase Return

Return defective items to your supplier

What Is a Purchase Return?

A Purchase Return occurs when you send items back to your supplier — for example, because they arrived defective, broken, or were the wrong item. Unlike a Sales Return, this decreases your stock.

Processing a Purchase Return

1	Click 'Purchase Return' in the navigation menu.
2	Select the product you are returning to the supplier.
3	Check the Current Stock shown in the information panel.
4	Enter the Return Quantity. This cannot exceed the current stock.
5	Enter the Return Reason (e.g., 'Defective batch', 'Wrong item received', 'Damaged packaging').
6	Click 'Return to Supplier'. Stock is reduced by the entered quantity.

Important: You cannot return more units than you currently have in stock. If the quantity exceeds stock, the system will block the action and show an error.

What Gets Recorded

- A stock log entry is created with action type 'Purchase Return'.
- The reason and quantity are saved for supplier dispute resolution if needed.
- Stock is reduced, reflecting that the items are no longer in your possession.

Module 7

✕Product Loss

Record inventory lost due to damage, expiry, or spoilage

What Is a Product Loss?

Product Loss is used to remove items from inventory that can no longer be sold — such as expired goods, items damaged in storage, or broken merchandise. This action permanently reduces stock and cannot be undone.

Recording a Product Loss

1	Click 'Product Loss' in the navigation menu.
2	Select the affected product from the dropdown.
3	Review the current stock level in the product information panel.
4	Enter the Loss Quantity — how many units are being written off.
5	Enter the Loss Reason — be specific (e.g., 'Expired March 2026', 'Damaged by flood', 'Broken during handling').
6	Click 'Record Loss'. A confirmation dialog appears listing the product, quantity, and reason.
7	Click Yes to confirm. The stock is permanently reduced.

Warning: This action is irreversible. Once confirmed, the stock reduction and log entry cannot be undone. Always double-check the product and quantity before confirming.

What Gets Recorded

- A stock log entry is created with action type 'Loss'.
- The reason is saved for inventory audit purposes.
- Both the stock before and after are recorded.

Common Loss Reasons: Expired, Damaged, Broken, Spoiled, Stolen, Water Damage — use specific reasons for better record-keeping.

Module 8

Reports

Analyze sales performance, profit, and stock activity

Overview

The Reports module provides a comprehensive view of your business performance. It includes summary statistics, visual charts, a detailed profit breakdown per product, and recent activity logs. All data can be filtered by date range and exported to PDF.

Setting the Date Range

At the top of the Reports page, use the From and To date pickers to define the period you want to analyze.

- Click 'Apply Filter' to refresh all charts and statistics for that date range.
- Click 'Reset' to return to the default range (all-time data).

> **Tip:** If the selected date range has no sales data, the system automatically falls back to showing all available data so you are never left with a blank report.

Summary Statistics Cards

Four cards at the top show the key metrics for the selected period:

Card	What It Shows
Total Revenue	The total amount earned from all sales in the period.
Total Cost	Total cost of goods sold (quantity × unit cost per product).
Net Profit	Revenue minus Cost. Green if positive, red if negative.
Profit Margin	Net Profit as a percentage of Total Revenue.

Performance Charts

Below the summary cards is an interactive bar chart. Use the Chart Type dropdown to switch between three views:

- Revenue — Top 10 products ranked by total sales revenue.
- Profit — Top 10 products ranked by net profit earned.
- Units Sold — Top 10 products ranked by total quantity sold.

 Tip: Switch between chart types to get different perspectives on your best-performing products. A product might rank high in revenue but lower in profit if its unit cost is high.

Product Profit Analysis Table

This table shows a row for every product that had sales in the selected period. Columns include:

- Product — The product name.
- Units Sold — Total quantity sold across all transactions.
- Revenue — Total sales income from this product.
- Cost — Total cost of goods sold for this product (quantity × unit cost).
- Profit — Revenue minus Cost. Displayed in green (positive) or red (negative).
- Margin % — Profit as a percentage of Revenue.

 Note: Profit figures depend on an accurate Unit Cost being set on each product. If Unit Cost is 0 or not set, the profit for that product will appear artificially inflated.

Recent Sales & Stock Activity

At the bottom of the Reports page are two side-by-side tables showing the 10 most recent sales transactions and the 10 most recent stock log entries. These give you a quick at-a-glance view of recent activity. For a full searchable history of every action, use the Audit Trail module.

Exporting to PDF

1	Set your desired date range and click 'Apply Filter' if needed.
2	Click the green 'Export to PDF' button in the top-right corner of the Reports page.
3	A Save dialog opens. Choose a location and filename, then click Save.
4	The PDF is generated automatically and a confirmation message appears with the file path.
5	Open the PDF file from the saved location.

The exported PDF includes: a professional header, the report date range, all four summary statistics, the Product Profit Analysis table (up to 20 products), recent sales transactions (up to 15), recent stock activity (up to 15), and a generation timestamp in the footer.

 Tip: PDF filenames are automatically timestamped (e.g., Sales_Report_20260218_143022.pdf) so multiple exports do not overwrite each other.

Module

Cycle Count

Compare physical stock against system records and identify variances

What Is a Cycle Count?

A Cycle Count is a physical inventory check where you count items on the shelf and compare them to what the system expects. Variances reveal shrinkage, data entry errors, or unrecorded losses. Each count is organized into a Session so results are stored and can be reviewed at any time.

Starting a New Count Session

1	Navigate to 'Cycle Count' from the left menu.
2	Optionally enter session notes (e.g., 'End of month count') in the Notes field on the left panel.
3	Click '+ Start New Count'.
4	A confirmation dialog shows how many products will be included. Click Yes to confirm.
5	The new session is created and automatically selected. The count sheet appears on the right.

Tip: When a session starts, the system records the current stock quantity of every product as the 'Expected' amount. Changes made to stock after the session starts do not affect the expected quantities.

Entering Physical Counts

1	In the Count Sheet table on the right, click any row to select a product.
2	The selected product's name appears at the top of the entry panel.
3	Enter the Physical Count (the actual quantity you counted on the shelf).
4	Optionally add a note (e.g., 'Found in back storage').
5	Click 'Save Count'. The variance is calculated automatically and the next uncounted item is selected.

Understanding Variance Status

Each item in the count sheet is assigned a status badge:

Status	Color	Meaning
Pending	Orange	Item has not been counted yet.
OK	Green	Counted quantity matches the expected quantity exactly.
Overage	Blue	Counted quantity is higher than expected (more items found than recorded).
Shortage	Red	Counted quantity is lower than expected (fewer items found than recorded).

The summary cards at the top of the session view show: Total Items, Counted, Pending, and Variances (with total variance value in ₱). These update live as you save each count.

Completing a Session

1	Count as many items as needed. You do not have to count all items before completing.
2	Click 'Complete Session' when done.
3	If there are uncounted items, a warning dialog will ask if you want to complete anyway. Click Yes to proceed.
4	The session status changes to 'Completed' and becomes read-only. It remains visible in the Sessions list for future reference.

 Note: Completing a session does NOT automatically adjust system stock to match physical counts. You must manually use Restock, Sales Return, Purchase Return, or Product Loss to correct any variances.

Module 10

Audit Trail

Complete log of every action performed in the system

What Is the Audit Trail?

The Audit Trail is a complete, append-only log of every meaningful action performed in the system. Every time a product is added, edited, or deleted; a sale is recorded; stock is adjusted; a cycle count is performed; or a backup is created — a timestamped entry is written to the audit log. This provides full accountability and makes it easy to trace any change.

Viewing & Filtering Entries

Navigate to 'Audit Trail' from the left menu. All entries are displayed in a table sorted by most recent first. You can filter the list using any combination of the following:

Filter	How It Works
Search	Type any keyword to filter by product name, details, or action type in real time.
Action Type	Use the dropdown to filter by a specific action (e.g., show only 'Sale' or only 'Product Deleted').
From / To Date	Restrict results to a specific date range.
Refresh	Reloads all entries from the data file — useful if multiple users may be adding data.
Clear	Resets all filters back to default.

The record count bar shows 'Showing X of Y records' so you always know how many results match your current filters.

Audit Action Types

The Audit Trail records the following action types:

Action	Triggered By
Product Added	A new product was saved.
Product Edited	An existing product's details were changed. Old and new values are recorded.

Product Deleted	A product was permanently deleted.
Sale	A sales transaction was completed.
Restock	Stock was added to a product.
Sales Return	A customer returned an item.
Purchase Return	An item was returned to the supplier.
Product Loss	A product loss was recorded.
Cycle Count Started	A new cycle count session was created.
Cycle Count Item	A physical count was saved for one item in a session.
Cycle Count Completed	A cycle count session was marked complete.
Manual Backup	The user clicked 'Backup Now'.
Auto Backup	The system automatically created a scheduled backup.

 Tip: For product edits, the Old Value and New Value columns show the before and after state in pipe-delimited format (e.g., Name:OldName|Price:50.00). This makes it easy to see exactly what changed.

Module 11

Backup

Automatic weekly backups with manual trigger option

Auto Backup

The system automatically backs up all data files every 7 days. On application startup, it checks whether a backup is overdue. If 7 or more days have passed since the last backup, a backup runs immediately. After that, the system checks every hour and backs up if the schedule is due.

Each backup is a ZIP file saved to the Data/Backups/ folder inside the application directory. It contains all Excel data files (Products.xlsx, Sales.xlsx, StockLogs.xlsx, AuditTrail.xlsx, CycleCounts.xlsx) plus a backup_manifest.txt with metadata.

> **Tip:** The system keeps a maximum of 8 backups (approximately 2 months of weekly history). When a new backup is created and there are already 8, the oldest is automatically deleted.

Manual Backup

1	Navigate to 'Backup' from the left menu under the SYSTEM group.
2	Review the info cards: Last Backup date, Next Scheduled Backup date, and how many backups are currently stored.
3	Click 'Backup Now' to create an immediate backup.
4	A success message confirms the backup was created and shows the filename.
5	The backup is logged in the Audit Trail as 'Manual Backup'.

> **Tip:** You can click 'Open Backup Folder' to open Windows Explorer directly to the Backups folder, making it easy to copy files to an external drive or cloud storage.

Backup History

The Backup History table at the bottom of the Backup page lists all stored backups in reverse chronological order (newest first). For each backup you can see: the filename, when it was created, and its file size in KB. The full file path is also shown so you can locate the file manually if needed.

> **Note:** Backups are stored locally in the application's Data/Backups/ folder. For best protection, periodically copy backup ZIP files to an external location such as a USB drive or cloud storage service.

Quick Reference

Navigation Menu Summary

Menu Item	Action	Effect on Stock
□ Products	View product catalog	None — read only
➕ Add/Edit Product	Create or modify products	Sets initial stock on creation
□ Sell Product	Record a sale	Decreases stock
□ Restock	Add incoming inventory	Increases stock
□ Sales Return	Customer returns item	Increases stock
□ Purchase Return	Return item to supplier	Decreases stock
✕ Product Loss	Write off damaged/expired goods	Decreases stock
□ Reports	View analytics & export PDF	None — read only
□ Cycle Count	Physical inventory count with variance tracking	None — read only
□ Audit Trail	View complete history of all actions	None — read only
□ Backup	View backup status and trigger manual backup	None — read only

Common Questions

Why is my profit showing as ₱0 or inflated?

The profit calculation relies on the Unit Cost field in each product. If Unit Cost is 0 or blank, the system calculates zero cost, making profit equal to revenue. Make sure to set an accurate Unit Cost when adding or editing products.

Why does the report show all-time data instead of my selected date range?

This happens when no sales exist within the selected date range. The system falls back to showing all data to avoid a blank report. Check that your date range includes dates when sales were recorded.

Can I recover a deleted product?

No. Deleting a product permanently removes it from the product list. However, past sales and stock log entries that mention the product are still retained in the data files for record-keeping. The deletion is also logged in the Audit Trail.

Can I undo a Product Loss?

No. Product Loss is a permanent action and cannot be reversed within the system. Always confirm the quantity carefully before accepting the confirmation dialog. The loss is permanently recorded in the Audit Trail for your records.

My Excel data file is locked and changes aren't saving.

This happens when another program (such as Microsoft Excel) has the data file open at the same time. Close any open Excel windows and try the operation again.

Does completing a Cycle Count adjust my stock automatically?

No. Completing a cycle count session records the variances but does not automatically update stock. To correct variances you must manually use Restock (for overages you want to add), Sales Return (to bring in missing stock), or Product Loss (to write off shortages).

How do I restore from a backup?

Navigate to 'Backup' and click 'Open Backup Folder'. Locate the ZIP file for the date you want to restore. Stop the application, then extract the ZIP and copy the .xlsx files into the Data/ folder inside the application directory, replacing the existing files. Restart the application.

Stock Log Action Types

Every stock change creates a log entry with one of these action types:

Action	Meaning
New Product	A new product was added to the system.
Sale	A sale was completed and stock was deducted.
Restock	Stock was added from incoming inventory.
Sales Return	A customer returned an item; stock increased.
Purchase Return	An item was returned to the supplier; stock decreased.
Loss	Items were written off due to damage, expiry, or spoilage.